

Team Condor Design, Analysis, and Flight Performance Report

Dustin Feldman (35070323), Mikayla Holden (34990747), Apoorvee Dahiya (35154519), Alec Stanley (33963975), Dean Neaves (33118140)

MAE 2404, Department of Mechanical and Aerospace Engineering
Monash University, Clayton, Victoria, 3800 AUSTRALIA

Abstract

This project focused on the design, construction, and progressive refinement of a small glider optimised for a low Reynolds number flight between approximately 34,000 and 85,000. Through three major design iterations, the aircraft evolved from a structurally weak and aerodynamically inefficient prototype into a stable, lightweight configuration featuring an Eppler E423 airfoil and optimised aspect ratio of 8. Structural efficiency and aerodynamic smoothness were improved through modifications in relation to material and structure, weight reduction, and surface refinement. Final testing demonstrated stable flight characteristics and controlled glider performance. Key learnings included the critical need for a balance between stiffness and weight, the sensitivity of a low Reynolds flight to surface quality and geometry, as well as the importance of rapid prototyping in advancing toward a structurally robust and aerodynamically stable design.

Table of Contents

1.0 Introduction.....	1
2.0 Design Process.....	2
2.1 Preliminary Literature Research/Review.....	2
2.1.1 Wings.....	2
2.1.2 Fuselage.....	2
2.1.3 Tail.....	3
2.1.4 Center of Gravity/Mass/Pressure.....	3
2.1.5 Materials and Manufacturing.....	3
2.2.1 Key Learnings.....	3
2.4 Model 3.....	5
2.4.1 Drastic Mass Reduction.....	6
2.4.2 Adjustability [For stabilization], and durability [for achieving predictable flights across many flights].....	6
2.5 Post Model 3 Improvements and Reflection.....	6
2.5.2 Design Process.....	7
3.0 Performance Predictions.....	8
3.1 Definitions of Constants.....	8
3.2 Flow Regime & Lift.....	9
3.3 Drag & Glide Ratio.....	10
3.4 Stability & Moments.....	12
4.0 Flight Performance.....	13
4.1 Flight Testing & Model Alterations.....	13
4.1.1 Identifying Optimal Release.....	13
4.1.2 Model 1.....	13
4.1.3 Model 2.....	14
4.1.4 Model 3.....	14
4.2 Live Flight Performance.....	14
4.2.1 Overall Performance.....	14
4.2.2 Performance Comparison.....	15
5.0 Conclusion and Recommendations.....	16
6.0 References.....	17
7.0 Appendices.....	18
Appendix A: Flight Prediction Calculations Code (MATLAB).....	18
Appendix B: Wing Loading Glider Velocity Formula.....	19
Appendix C: Lift Coefficient For Different Angles of Attack and Cambers, and Diagram of The Turbulent Wake at a High Angle of Attack.....	19
Appendix D: Diagram of Wingtip Vortices, and Graphs of The Aspect Ratio Performance Comparison For NACA 2414 Aerofoil.....	20
Appendix E: Oswald Span Efficiency Factor, and Different Wing Distributions.....	22

1.0 Introduction

The pursuit of sustained, efficient gliding flight represents a fundamental challenge in aerodynamics as it requires a careful balance between lift production and the minimisation of the effects of induced and parasitic drag. For an unpowered glider in steady flight, the lift force must be able to counteract its weight, while the drag force dictates the rate of descent. The efficiency of this process is quantified by the lift-to-drag ratio (L/D), or glide ratio, which defines the horizontal distance an aircraft can travel forward for each unit of altitude lost (Anderson, 2011). The primary object in designing aircraft for endurance and range is the maximisation of this ratio hence making it the critical performance metric for this project.

The principles of glider design rely on the maintenance of a force equilibrium in steady flight. As shown in Figure 1, the glide path angle, γ , directly links the aerodynamic forces to the aircraft's kinematics. In this equilibrium state, the ratio of lift to drag is equivalent to the ratio of horizontal velocity to vertical sink rate, and consequently, the ratio of horizontal distance covered (R) to height lost (h). This principle allows for the performance of a glider to be predicted analytically and validated experimentally through simple flight distance measurements.

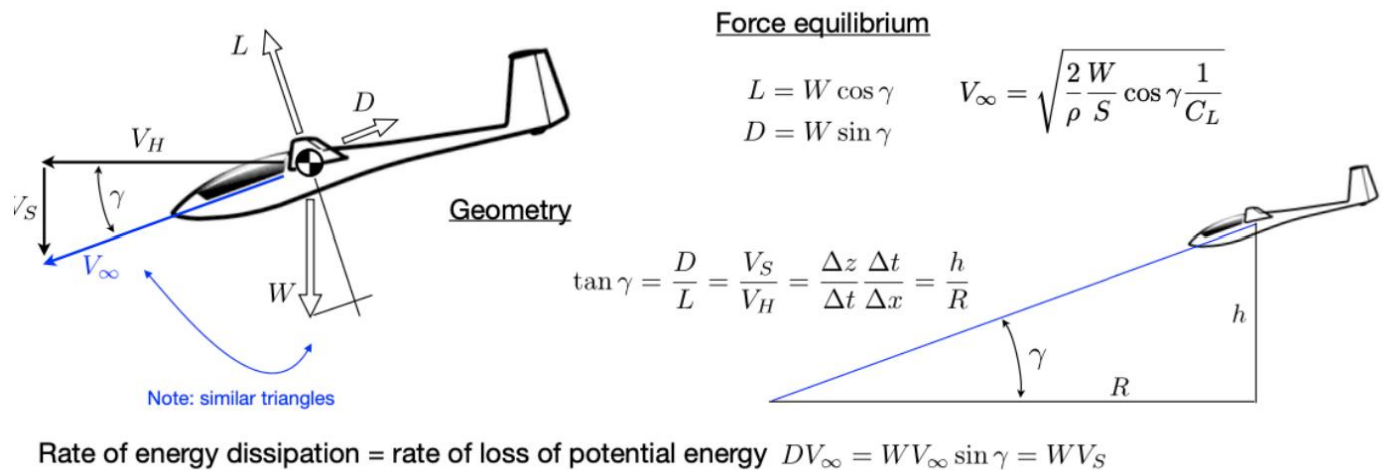


Figure 1: Representation of gliding flight depicting how lift and drag relate to vertical and horizontal motion

The aim of this project is to design, construct, and test a hand launched glider that achieves a sustained glide and demonstrates a maximum lift-to-drag ratio. The design process is constrained by a maximum wingspan of 1.0 metre, a standing height release, and the prohibition of any active control mechanisms or propulsive systems. To navigate these constraints, the design process applied core aerodynamic principles. Thin

Aerofoil Theory and Lifting-Line Theory were used to predict the performance of the wing, influencing the selection of a high lift airfoil section that is appropriate for the low Reynolds number range (approximately $10^4 - 10^5$) observed in small scale models (UIUC Applied Aerodynamics Group, n.d.). Furthermore, the design focused on static stability by ensuring the centre of gravity was positioned forward of the aerodynamic centre, a critical factor for stable flight (Boldmethod, 2016).

The following report details the complete design process, from initial concepts to final flight demonstration. It will cover the justification for the selected configuration, the application of aerodynamic theory for performance prediction, and an analysis of the live flight results, leading to a comparison of the attained glide ratio with theoretical values and the performance of peer designs.

2.0 Design Process

Condor's design process is governed by two main design philosophies. The first embracing failure; prototyping quickly, failing, reflecting and adapting with direction addressing the failures/limitations of the previous design. The second is consulting literature; ensuring our approaches are backed by literature, particularly relating to consolidating content covered in Aerodynamics 1.

2.1 Preliminary Literature Research/Review

To encourage reasoned design decisions that build on existing literature, the Condor team began with delegating preliminary research areas of wings, fuselage, tail, center of gravity, mass and pressure and materials/manufacture.

2.1.1 Wings

Airfoils should be selected based on shapes optimised for the relatively low Reynolds numbers (Re) of the glider. Even for low Re , the lift to drag ratio can be improved by over 30% by minimising parasitic drag and flow separation. Minimising Wing loading as defined by glider mass/wing area, will ensure the glider can fly at lower speeds without stalling (Appendix B). The wing's angle of attack should be set around 8 to 10 degrees, as accounting for airfoil zero-lift angles, this number is found to maximise lift generation, before flow separation effects emerge, causing turbulent wakes which undesirably cause significant drag increases and instability (Appendix C).

The effects of Wing-tip vortices on finite wings should be considered, as they cause downwash, creating induced drag which inhibits lift by reducing effective angle of attack. Adopting higher aspect ratios (AR) can minimise these effects, exhibiting closer behaviour to infinite wings. However, when considering real world manufacturing, it must also be considered that for higher AR longer, skinnier wings can be more flimsy and have more rotational inertia, also considering constraint 'D1.5' fixed span, higher aspect ratios result in less total wing area, so potentially increased wing loading (Appendix D). Wing shape may also be considered, through adopting elliptical [optimal] or tapered wings, which is driven by Oswald span efficiency factors, While implementing these may slightly increase L/D ratios, the design complexity is likely heavily increased.

Finally compared to foldover airfoil, adopting more specific airfoil profiles can significantly enhance glide ratios, however later designs showed this is a major caveat, as if this profile introduces imperfections this can completely diminish the benefits of this (Appendix E).

2.1.2 Fuselage

The design of the fuselage should primarily focus on decreasing fuselage drag, as the fuselage only offers minor lift contributions. It should also be as smooth and streamlined as possible, including smoother transitions between other body parts. While nosecone shapes are less significant at such low speeds, technically sharper profiles are preferred. To minimise fuselage drag, the fuselage cross sectional area relative to the free-stream flow should be minimised, with a fineness ratio (length to max width) of 1:6 to 1:8. Tapered tailcones are suggested. and a tadpole fuselage profile is said to be best for gliders, but may be difficult to manufacture given low budget.

2.1.3 Tail

Conventional tail configuration is best fit due to its balance of simplicity, robustness and predictable handling. Tail surfaces should be sized using established tail volume coefficients to ensure adequate stability.

To guarantee pitch stability, the centre of gravity must be positioned forward of the aerodynamic centre. The design should prioritise a long tail moment arm in order to reduce the required tail area and minimum drag.

2.1.4 Center of Gravity/Mass/Pressure

The aerodynamic center where aerodynamic moment is constant irrespective of angle of attack is located typically 25 percent of the chord length from leading edge, the center of gravity should be placed in front of the aerodynamic center to obtain a positive static margin, it is important to have an optimised small positive static margin as too large may cause nose diving or otherwise limit L/D ratio. The center of gravity can be controlled via adding ballast as appropriate, generally through adding ballast in the form of steel nuts. Also a tailplane should be present to counter the downward gravitational force and stabilise the plane by generating a downward lift. It should make up 20-30 percent of total wing area.

2.1.5 Materials and Manufacturing

3 manufacturing options for the wing can be considered. Firstly, a foldover foam [sheet] wing, with internal spaces to shape the profile, offering quick manufacturing but imprecise airfoil, and potentially reduced strength. Secondly, a laser-cut or 3D-printed airfoil profile coupled with a hotwire cutter into foam, which would enable a precise airfoil profile, with rigid surfaces, and lighter wing, but potentially incurring high cost [out of budget] to obtain a wire cutter.

Finally, creating many laser-cut or 3D printed ribs, and scaffolding them, and then wrapping them in a specialised polyester wrap, but otherwise can use paper or tape. It is positive in its ability to create a precise airfoil shape, without sourcing hotwire cutters, but difficult to maintain/fix, and risky if wrap is imperfect. In terms of other manufacturing considerations it is suggested to minimise skin friction with low surface wraps/paint, and use a solid extrusion structure for the fuselage to enable stability modifications.

2.2 Model 1

Model 1 intends to closely adhere to the “quickly build, fail, reflect, improve” philosophy. It takes the simplest foldover foam manufacturing, at a modest 6 AR, [internally reinforced with an 80g 10mm hollow aluminum extrusion of 1m], a sturdy fuselage with a rough tail is incorporated. It uses tape to position and connect the half-wings and tail, with the hope that both could be dynamically adjusted for stability.

2.2.1 Key Learnings

The pitching and yawing motion as displayed in Figure 2, suggests the need for stronger wing-fuselage connection, and larger vertical tail sizes. In its ‘rough-draft’ nature Model 1 also demonstrates the importance of precise control of variables in relation to literature, so variables can be isolated and optimised. This could be achieved through attempting other wing manufacturing methods, and being more analytical in designing wing and tail sizes, geometry and placement.



Figure 2: Image of Model 1 in Flight

2.3 Model 2

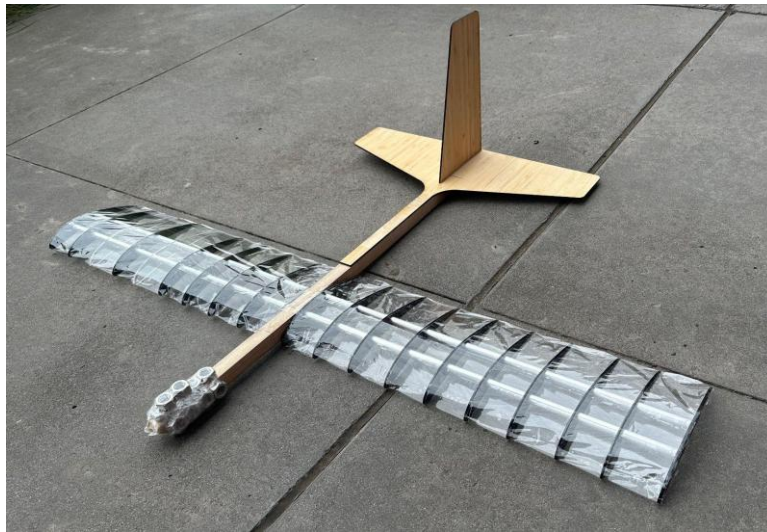


Figure 3: Image depicting Model 2

Model 2 centres around applying literature with greater precision, to control and optimise variables. A preliminary Reynolds number estimation was taken as ranging from 34,000 to 85,000, based on standard sea level assumptions and limited flight velocity understanding. A number of low Reynolds number optimised airfoil profiles were considered, such as the Selig 1223, the Eppler E387. However the Eppler E423 was selected, as it had a greater camber for similar performance, and the camber profile was essential, as our planned manufacturing method involved using 20 lasercut airfoil ribs, internally connected by 210 mm aluminium cylindrical rods, and wrapped in packing tape and greater camber profile meant tape is less likely to stick together and tarnish performance, and the aluminium tube holes can have offset on both sides to minimise risk of cracking.

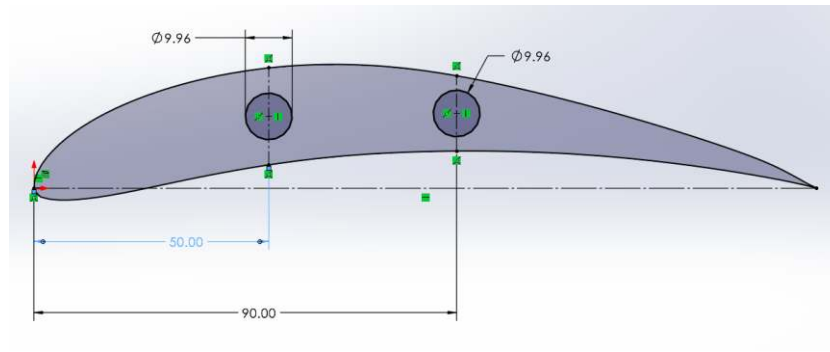


Figure 4: CAD sketch of aerofoil section

The airfoil support ribs were designed using Solidworks 2025 (Figure 4), downloading the E423 profile, and applying a spline, then prototyping the hole size to achieve snug but moveable placement on the aluminum support. The hole placement erroneously prioritised hole placements for minimising rib rotation, but this meant an extremely significant portion of weight contributed to a center of gravity behind the center of pressure, and this was exacerbated by the extremely heavy 3mm lasercut hardwood tail, which meant a total of 10 aluminium nuts had to be added to the front, which still was unable to achieve a positive static stability margin.

The tail was hard wood laser cut with an approximately 0.048m^2 tapered horizontal area, and a 0.024 tapered vertical, matching a 20% tail area to wing area ratio as dictated by literature, and connected via a thin section with superglue to the 20mm balsa fuselage. However this was found to snap easily, and be too heavy. The fact the aluminium supports were drilled to the fuselage, meant after usage they weaken, loosen, and prevent stable flight.

2.4 Model 3

Model 3 (Figure 5) reflects on Model 2, and makes drastic improvements based on Model 2's two identified critical failure points.



Figure 5: Image of Model 3

2.4.1 Drastic Mass Reduction

Model 3 now uses a single 90 degree aluminum angle bracket which is more effective at constraining airfoil-support rotation, and the fact the spar is located at the quarter chord, means it better supports the center of pressure, and drastically reduces the amount of required nose ballast. a AR of 8 was adopted which lowered the required tail sizes, the tails were now made of cut corflute, achieving sufficient rigidity, at a lower mass.

2.4.2 Stabilisation, Reliability, and Durability

A 3D printed tail mount (Figure 5) is now used allowing greater surface area for adhesion and impact, ensuring on impact, less force is concentrated. The 7.5 cm bottom connection enhances impact resistance, and provides dynamic adjustability of tail location without removing glue. A 3D printed Wing- fuselage joint, sturdily constrains the wing perpendicular to the fuselage , preventing yaw, and drastically enhances durability as there is no direct contact between the wing support and fuselage, it also offers a similar adjustability along the fuselage for stability. It was designed at an AOA of 1.2 Finally a 3D printed streamlined nose was added to support the fuselage during Nose Dive impacts, and act as a more streamlined and fixed ballast than nuts.



Figure 5: Model 3 Tail Joint (right), Wing-Fuselage-Joint (centre), and Nosecone-Ballst (left)

2.5 Post Model 3 Improvements and Reflection

2.5.1 Material, Manufacturing and design

The Team's thorough research, identified and applied design decisions, which based on the content of Aerodynamic 1 would strongly optimise L/D and performance in just about all areas, such as optimal wing shape, fuselage length and cross section, tail area placement, and stability calculations. This could be effectively optimised for within a considerably higher budget through sourcing more expensive, and vastly superior materials and manufacture techniques, like a high tech polyester controlled adhesive shrinking system for wrapping hot wire foam cutting, and carbon fibre spars. However, in the context of a very low budget glider, adopting these design decisions led to excess mass, and the imperfections associated with the tape would have diminished the benefit of using an optimised airfoil profile such as the E423. In the future, low budget gliders should consider returning to foldover foam, as the most essential part can be obtaining a rigid airfoil profile.

Other Design Modifications include increasing inbuilt angle of attack of wing-fuselage joint, to account for effective angle of attack reductions due to real world effects such as induced drag, or manufacturing imperfections. Enhanced grips and further stability tuning should be adopted in future flight.

2.5.2 Design Process

In hindsight, instead of delegating an individual to manufacturing methods for each part, instead the researchers of that part should be responsible for researching manufacturing of their part to ensure literature idealised designs are balanced with manufacturing considerations. An additional person should be dedicated to wings and their associated variables to ensure all aspects are thoroughly accounted for in design.

3.0 Performance Predictions

3.1 Definitions of Constants

To theoretically determine the performance levels of our proposed glider, we must first make certain assumptions, and define fixed environmental parameters. We know that quantities such as air density and temperature vary at different altitudes, however since the flight does not occur over a large vertical distance, we can safely assume quantities such as these to be constant.

The table below summarises a list of environmental quantities, and the values that are set for them in our calculations:

Table 1: Summary of environmental quantities, and constant values used in calculations

<u>QUANTITY</u>	<u>SYMBOL</u>	<u>VALUE</u>
Air Density	ρ	1.225 kg/m ³
Temperature	T	293 K
Dynamic Viscosity (air)	μ	1.8e-5 Pa.s
Sound speed (air)	A	343 m/s

Alongside these environmental constants, we also must define values specific to our glider. The airfoil section that these calculations have been based on is the Eppler E423 cambered airfoil (Figure 6).

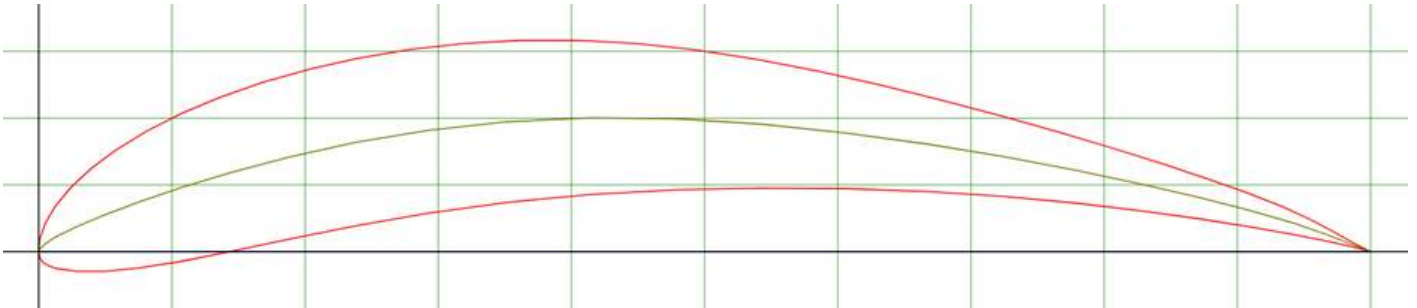


Figure 6: The airfoil profile of our final constructed glider, based on the Eppler E423 cambered airfoil

This airfoil has the dimensions and properties tabulated below (Table 2):

Table 2: Properties of the airfoil profile pictured above in Figure 6

<u>QUANTITY</u>	<u>SYMBOL</u>	<u>VALUE</u>
Wingspan	b	1 m
Aspect Ratio	AR	6
Chord Length	c	0.1667 m
Wing Area	S	0.1667 m ²
Glider Mass	m	300g

Span Efficiency	c	0.9
Zero Lift Angle	$\alpha_{L=0}$	-2.5 degrees

3.2 Flow Regime & Lift

Firstly, we require an estimate for the velocity that our glider will have during its gliding flight. To determine this, we can use the equation for the lift coefficient. During a glide, the downwards weight force must be balanced by the upward lift force generated by the glider. The lift coefficient formula, rearranged for velocity under these conditions, is given as;

$$V = \sqrt{\frac{2 * W}{\rho * S * C_L}} \quad [1]$$

where W is the weight force, S is the wing area, and C_L is the lift coefficient. We note that we can drop the cosine term from the balance, by approximating a small glide angle.

We still require an estimate for the lift coefficient, before we can determine the glide velocity however, which is obtained by the introduction of Lifting Line theory, giving us the relation:

$$C_L(\alpha) = a * (\alpha - \alpha_{L=0}) \quad [2]$$

Where we see that the lift coefficient is a function of the angle of attack of our wing. It is also subjected to the factor a, which is a lift slope given by:

$$a = \frac{a_0}{1 + \frac{a_0}{\pi AR}} \quad [3]$$

From Thin Aerofoil theory, we can approximate the term $a_0 \approx 2\pi$, and using our glider data provided in Table 2, we calculate a lift slope of $a = 4.7124$, which can be visualised in Figure 7 below.

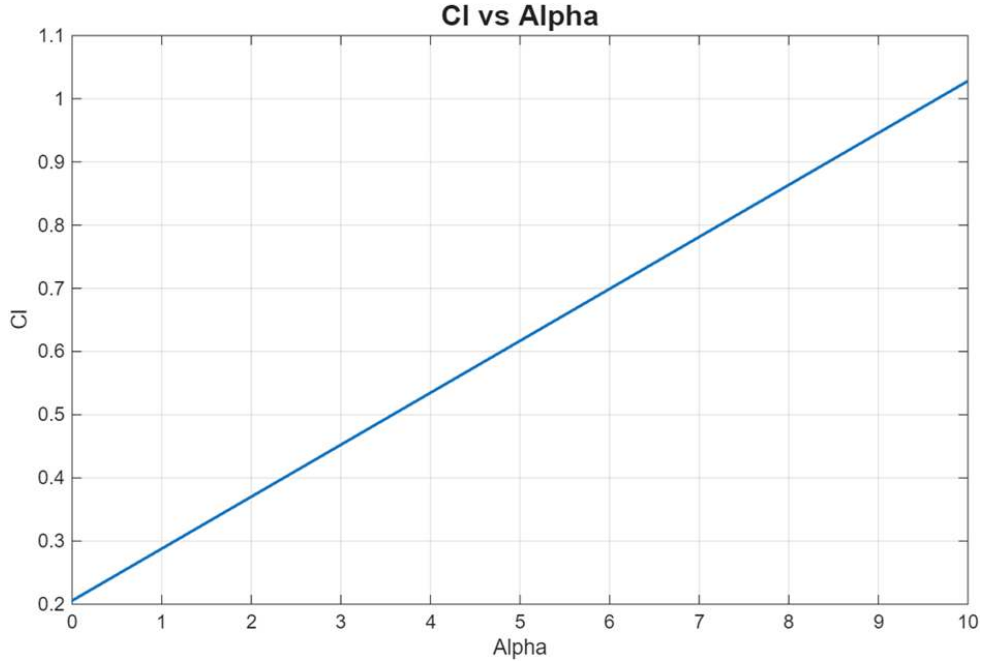


Figure 7: Line

displaying Coefficient of Lift (Cl) vs Angle of Attack (alpha)

Graph

We can now use these values to compute the glide velocity. Our glider uses an angle of attack of $\alpha = 1$ degree, and hence using this value in equations 1 and 2, we obtain a glide velocity of 10 m/s.

This is a crucial data point, as it now allows us to determine the Reynold's number and Mach number of the flight, which describe the airflow regime.

The Reynold's number and Mach number are subsequently found by:

$$Re = \frac{\rho V c}{\mu} \quad [4] \quad M = \frac{V}{A} \quad [5]$$

Giving values of $Re = 1.13e5$, and $M = 2.92e-2$.

3.3 Drag & Glide Ratio

Following lift, we must also determine the drag that our specific glider experiences during flight, to properly assess its glide ratio. To do this, we firstly define the total drag coefficient as:

$$C_D = C_{D0} + \frac{C_L^2}{\pi e AR} \quad [6]$$

where it has two components, a parasitic drag component, C_{D0} which acts as a constant drag, and an induced drag component, which is dependent on the lift produced by the wing.

To find the contribution of parasitic drag, we use a summation approximation of glider components as follows:

$$\sum C_f * Form Factor * \frac{exposed area}{wing area} \quad [7]$$

Where we see that the parasitic drag is dependent on the skin friction coefficient C_f , the components form factor, and finally the exposed area of the components, relative to the wing area.

To perform this approximation, we split the glider up into 3 components, the wing, the fuselage, and the tail. Beginning with the wing, we take a form factor of 1.15 (determined from the thickness ratio of E423), and an exposed area of $2 \cdot S$ (lower and upper surfaces). For the fuselage, we assume a form factor of 1.5 (determined from fineness ratio), and an exposed area of 0.064 m^2 according to the area of a rectangular beam. Finally for the tail, we take a form factor of 1.05 (flat piece of corflute), with an exposed area of 0.1 m^2 . We also require an approximation of the skin friction coefficient which we obtain from:

$$C_f = \frac{0.074}{Re^{\frac{1}{5}}} \quad [8]$$

Whereby using our calculated Reynold's number, we find a skin friction coefficient of $7.2 \text{e-}3$. We can then combine this data into equation 4, to give a final parasitic drag coefficient of $C_{D0} = 2.53 \text{e-}2$. We can now display visually the relation between the drag coefficient and the lift coefficient as below in Figure 8.

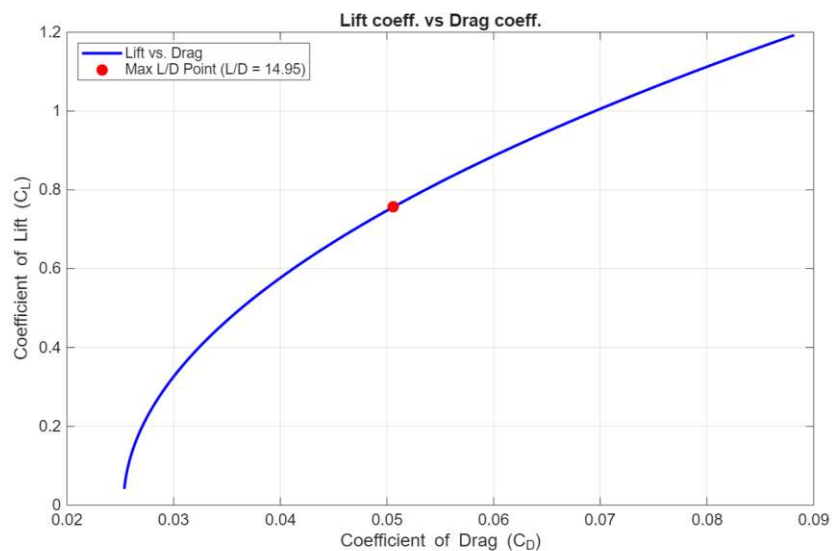


Figure 8: Line Graph displaying Coefficient of Lift (CL) vs Coefficient of Drag (CD)

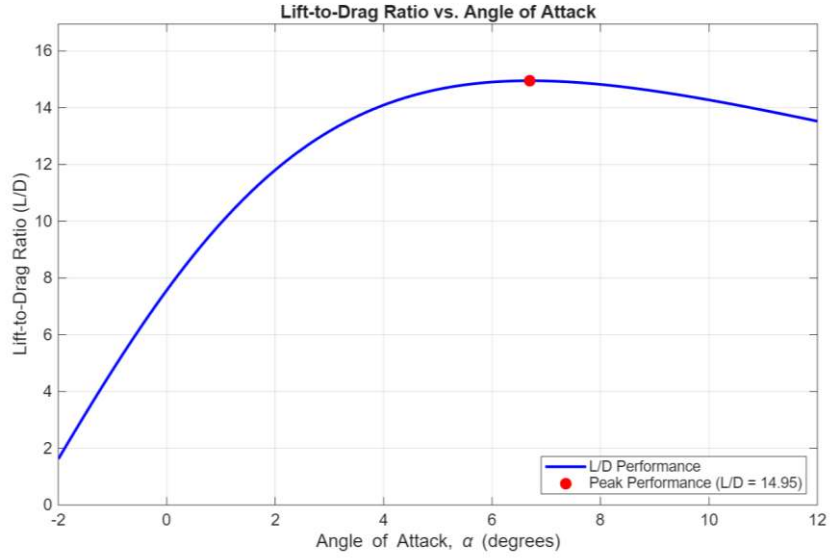


Figure 9: Line Graph displaying L/D ratio vs angle of attack

Finally, we can compute the glide ratio of the flight. The point at which the maximum glide ratio (L/D)

$$C_{L,Max\ glide} = \sqrt{C_{D0} * \pi e A R} \quad [9]$$

occurs is governed by the specific lift coefficient;

Which when computed gives a value of 0.757. This can then be used in the formula:

This gives a max glide ratio of L/D = 14.95. This value occurs at an angle of attack of 6.7 degrees with a lift

$$\frac{L}{D_{max}} = \frac{C_{L,Max\ glide}}{C_D} = \frac{C_{L,Max\ glide}}{C_{D0} + \frac{C_{L,Max\ glide}^2}{\pi e A R}} \quad [10]$$

coefficient of 0.757.

3.4 Stability & Moments

To determine the stability of the glider, we need to do a sum of moments about the aerodynamic centre. The aerodynamic centre is the point along the airfoil where pitching moment stays constant. The equation for the location of the aerodynamic centre for a cambered airfoil is given by:

$$x_{ac} = 0.25 - \frac{C_{m,0}}{C_{L,\alpha}}, \text{ where } C_{m,0} \text{ is the pitching moment coefficient at zero lift and } C_{L,\alpha} \text{ is the lift curve slope.}$$

This should line up with approximately 25% of the chord length. As calculated above, the lift curve slope is 4.7124. To determine the pitching moment for this airfoil at zero lift, we can run a quick simulation in the publicly available XFOIL program. After loading in the data file for the chosen airfoil, the program can run a series of tests at different angles of attack. The value for the pitching moment corresponding to a lift value closest to zero in the simulation is -0.08. This is a typical value for cambered airfoils, which usually lie between -0.15 and -0.05. Hence, $x_{ac} = 0.267$ the way down our airfoil from the leading edge.

With a chord length of 0.1667 metres, this corresponds to 4.45cm from the leading edge, or 34.45cm back from the front of the glider. Our goal is to show that the centre of gravity lies slightly closer to the leading edge than the aerodynamic centre. This gives us a positive static margin which causes stable flight.

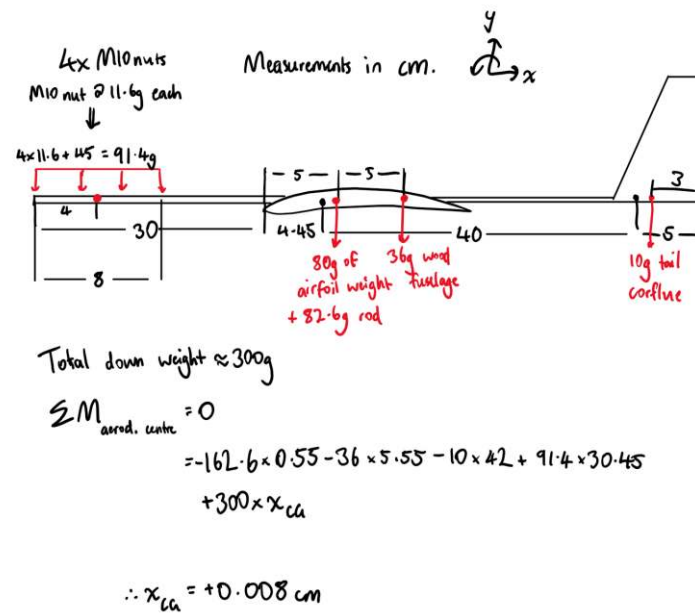


Figure 10: Calculations for the location of centre of gravity by taking moments about the aerodynamic centre.

As we can see, we have a very small positive static margin. However, this likely is not high enough to exhibit enough stability. We did notice this in our flight test, as the glider was quite susceptible to small perturbations that caused it to either dive or flip backwards. A more positive static margin would be beneficial for more stable flight, which could be achieved by moving the centre of gravity further towards the front.

4.0 Flight Performance

4.1 Flight Testing & Model Alterations

Over the two weeks spent building and refining the models, numerous test flights were performed to assess flight stability and compare results to theoretical predictions. These tests were crucial for tuning each model and improving overall performance. The flight testing was conducted on a large, relatively flat grassy area to minimise potential damage to the glider upon landing. However, being outdoors introduced challenges such as wind and other inconsistent conditions, which reduced the accuracy of comparisons between flights.

Before each flight, the plane was balanced to check that the centre of gravity was positioned ideally for stable flight. Achieving an appropriate centre of mass proved to be one of the greatest challenges across all models. After each test, adjustments were made by adding weights to shift the centre of mass into the ideal position for steady flight (Figure 11).



Figure 11: Image of our second model, with additional weights (hex nuts), taped onto the nose

4.1.1 Identifying Optimal Release

Establishing the optimal release method was essential for consistency and peak performance during the live flight demonstration. Before each flight, the glider was balanced to locate its centre of gravity along the fuselage, which was then used as the release point for throwing. As the flight constraints required a horizontal launch angle this parameter could not be changed, so ideal throwing force was important to determine, which was found through trial and error. Multiple flights were recorded under varying throwing forces while keeping other conditions as constant as possible. The footage was reviewed to identify which throw produced the greatest flight distance.

4.1.2 Model 1

The first test flight showed that the plane was tail-heavy, causing instability and leading the glider to pitch up. However, adding weights to the front made the glider tip unpredictably during flight as its behaviour was highly dependent on the throwing method, displaying significant instability. During testing, the tail and aerofoil components snapped (Figure 12), and the wings rotated about the central rod. These observations led

to the addition of a second parallel rod to keep the wings in place, and eventually led to a new model being created.

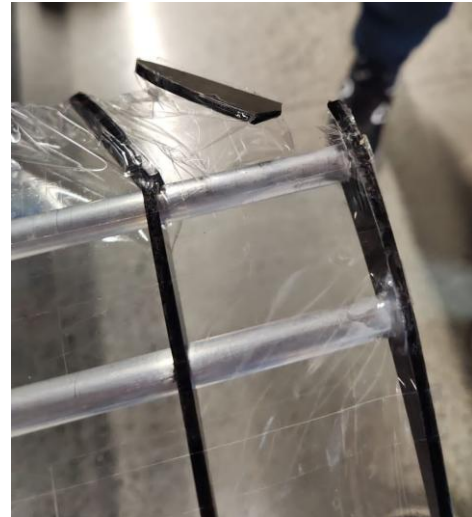


Figure 12: Image of where the tail component snapped off the fuselage (left), and the broken acrylic aerofoil sections (right) after the testflight of the second model

4.1.3 Model 2

Model 2 also began tail-heavy and pitched upwards, which was corrected by adding weights to the front through trial and error which made the flight become stable once adjusted. However, the overall weight of the glider was excessive. Although it flew smoothly, it descended quickly because the lift generated was insufficient to counteract its high weight. It was also found that the acrylic aerofoil sections frequently snapped, which ultimately led to the design of the final model.

4.1.4 Model 3

Model 3 demonstrated stable flight characteristics, with only some weights needing to be added to the nose of the fuselage. Several modifications were made to reduce weight, and build a more durable wing compared to model 2, as outlined in section 2.0 (design process). Only a small number of test flights were conducted to avoid potential damage before the live demonstration, and testing was mostly based on theoretical aspects.

4.2 Live Flight Performance

4.2.1 Overall Performance

During the live demonstration, the glider was released from an initial height of approximately 1.6 m for both flights. In the first flight, it travelled 5.3 m horizontally, while the second achieved a slightly longer distance of 7.3 m. From these results, the highest lift-to-drag ratio achieved was 4.56.

The variation between the two flights can be primarily attributed to differences in throwing force, as being able to maintain a consistent strength was difficult. This variability introduced some inconsistency in the outcomes as the first flight (5.3 m horizontal, 1.6 m vertical) yielded an L/D ratio of 3.31, while the second (7.3 m horizontal, 1.6 m vertical) achieved a slightly higher L/D ratio of 4.56. Between the first and second

flights, the vertical tail component also had to be readjusted and attached back on after snapping off during landing in the first test flight, which may have caused slight differences between flights. However, the more controlled environment of the indoor court led to less inconsistencies than the flight testing which was performed outdoors.

Aerodynamically, the flight path of the glider during the live demonstration suggests that the centre of gravity was positioned marginally too far back relative to the aerodynamic centre, as it initially pitched slightly upward before suddenly nose diving. This would have caused the nose to pitch upward at launch due to excess lift near the aerodynamic centre, followed by stall which caused the glider to dive as the airflow separated from the upper surface of the wing (Lewis, n.d.). This behaviour highlights the importance of balancing stability and lift generation in the overall glider performance.

4.2.2 Performance Comparison

Comparing these results to the theoretical calculations, it can be seen that the predicted L/D value was significantly higher (14.95) than the actual ratio achieved (4.56). This difference shows that although the design was theoretically capable of strong performance, real world factors limited the execution of how well it could perform. The main cause of this drop in performance came down to manufacturing and testing constraints. The looseness of the tape used along the wings during construction likely increased surface roughness and skin friction, heavily raising drag and reducing the actual lift-to-drag ratio. This uneven surface could also have caused small vortices and local flow separation along the wings, which are extremely detrimental to flight stability and overall aerodynamic efficiency. Furthermore, the relatively high weight of the glider coupled with a large experimental drag had a significant negative effect on its flight distance.

With the small number of test flights completed, the glider was never tuned to achieve a truly stable glide before the final launch. This limited how much we could adjust the centre of gravity to account for potential shifts in the centre of pressure, and it also meant that our throwing technique was not completely optimised. Without enough practice, and due to human inconsistency, each throw varied in strength and angle, which made achieving stable and repeatable flight much more difficult. Another factor that affected performance was the lack of inward force due to the high rigidity of our wings, which led to an additional lack of lateral stability. This made the glider extremely sensitive to small differences in the release, often causing it to rotate or roll unevenly in flight. These small instabilities quickly escalated mid flight, reducing both flight distance and consistency between attempts.

Overall, the surface imperfections, limited testing and optimisation, and design instabilities, explain why the achieved L/D ratio was much lower than the predicted value. The final glider performed relatively poorly compared to other teams, with the highest recorded L/D of any team being 12.47. The lowest single flight performance was 2.65, while the lowest result of our glider was 3.47, and out of the two flights, our glider achieved the best L/D ratio of 4.56, placing it 17th out of 19 teams.

5.0 Conclusion and Recommendations

This project revolved around the design and testing of a hand launched glider, resulting in a final model that achieved stable flight with a lift-to-drag ratio of 4.56. The design process, cycling through three prototypes, combined rapid prototyping with aerodynamic theory and resulted in a glider featuring an Eppler E423 airfoil and a 3D printed joint for stability purposes.

However, the performance of the glider was limited by significant manufacturing constraints. The use of heavy aluminium rods along with a taped skin wing contributed to an elevated drag and a less-than-optimal weight, which served as the primary reasons for the lower-than-predicted glide ratio. Insufficient flight testing time also prevented full optimisation of the centre of gravity and launch technique.

For a future prototype, the key improvements would be to use lighter materials like carbon fibre beams and a well-sealed wing to reduce weight and drag, and to allocate more time for systematic flight testing to refine stability and launch parameters.

6.0 References

Anderson, J. D. (2011). *Fundamentals of Aerodynamics* (5th ed.). McGraw-Hill Education.

Lewis, M. J. (n.d.). *Aircraft Stability & Control – Introduction to Aerospace Flight Vehicles*. Eagle Pubs. Retrieved October 27, 2025, from <https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/chapter/aircraft-stability-control/>

UIUC Applied Aerodynamics Group. (n.d.). *UIUC Low-Speed Aerofoil Tests*. https://m-selig.ae.illinois.edu/uiuc_lsar.html

What is Lift? (2022, July 21). Glenn Research Center. Retrieved October 27, 2025, from <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/what-is-lift/>

7.0 Appendices

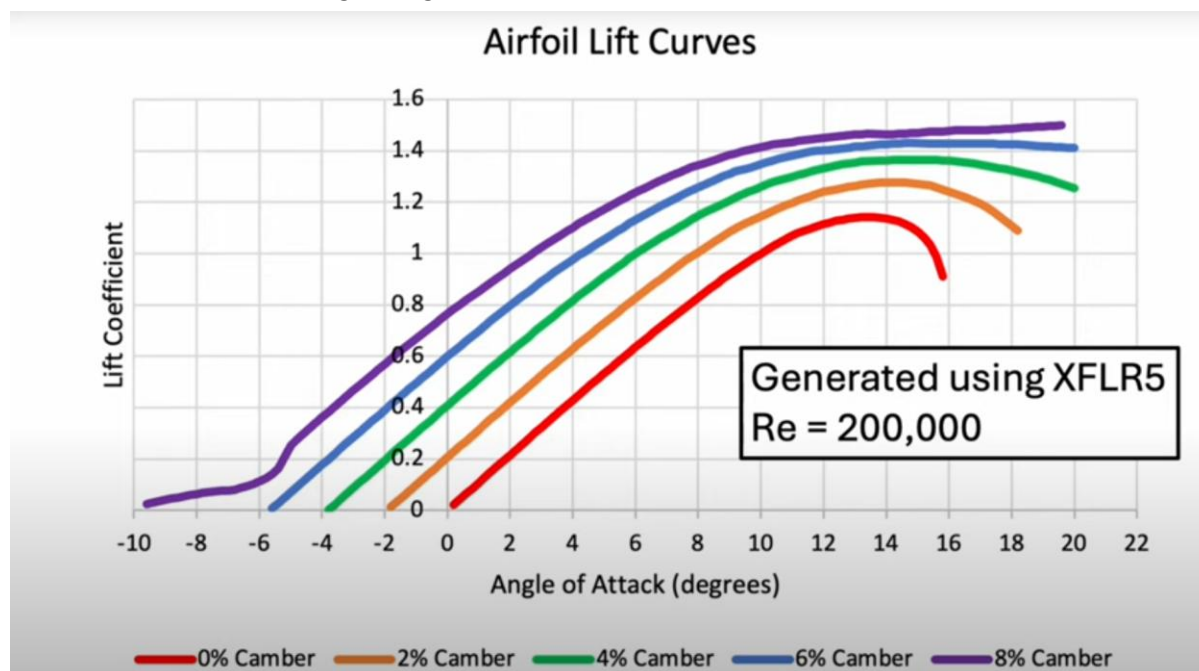
Appendix A: Flight Prediction Calculations Code (MATLAB)

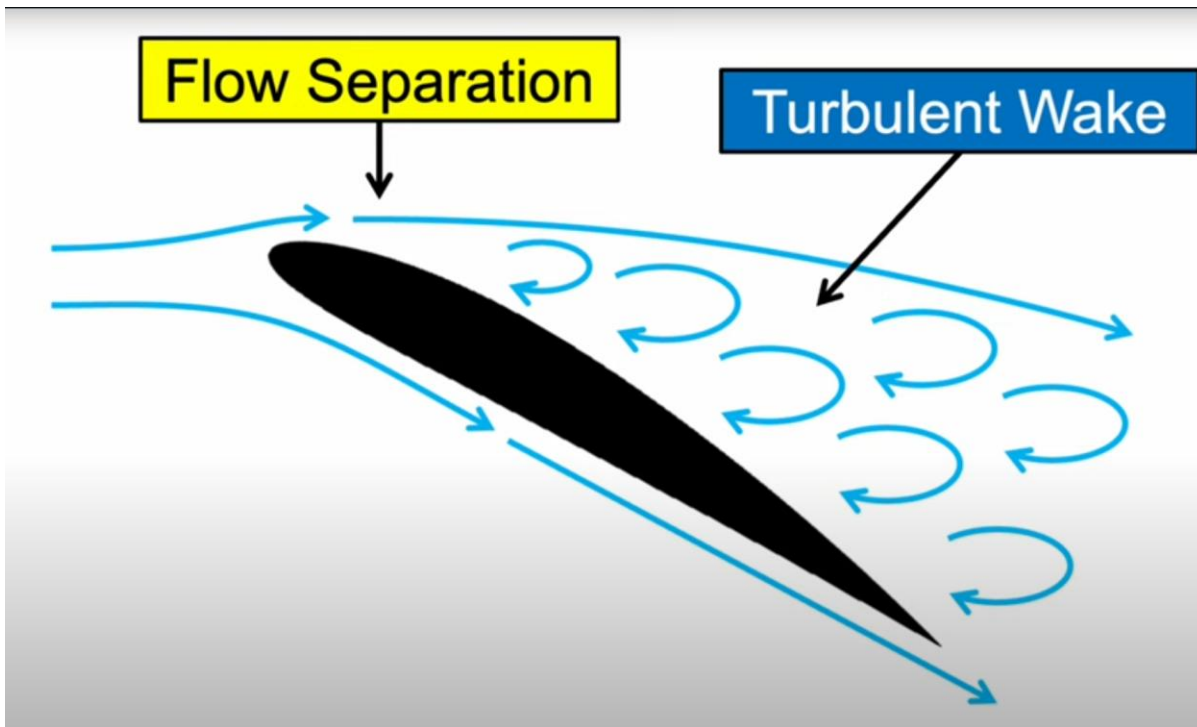
```
% Performance prediction plots
% Group CONDOR - by Alec Stanley
% Define glider and environmental constants
CD0 = 2.53e-2;      % Parasitic drag coefficient (calculated)
AR = 6;            % Aspect ratio
e = 0.9;           % Efficiency factor
a = 4.7124;        % Lift curve slope (per radian)
alphaL0_deg = -2.5;% Zero-lift angle of attack (degrees)
% Convert angles to radians for calculations
alphaL0_rad = deg2rad(alphaL0_deg);
% Generate data to calculate
alpha_deg = -2:0.1:12; % Range from -2 to 12 degrees
alpha_rad = deg2rad(alpha_deg); % Convert to radians
% Calculate CL, CD, and L/D for each angle of attack
CL = a * (alpha_rad - alphaL0_rad);
CD = CD0 + (CL.^2) / (pi * e * AR);
L_over_D = CL ./ CD;
% Find the maximum Lift-to-Drag ratio
[max_LD, index] = max(L_over_D);
alpha_max_LD = alpha_deg(index);
CL_max_LD = CL(index);
fprintf('Maximum L/D Ratio: %.2f\n', max_LD);
fprintf('Occurs at angle of attack: %.2f degrees\n', alpha_max_LD);
fprintf('Occurs at a lift coefficient (CL): %.3f\n', CL_max_LD);
% Generate Plots
% --- Plot 1: Coefficient of Lift vs. Coefficient of Drag ---
figure;
plot(CD, CL, 'b-', 'LineWidth', 2);
hold on;
plot(CD(index), CL_max_LD, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); % Mark max
L/D point
grid on;
title('Lift coeff. vs Drag coeff. ');
xlabel('Coefficient of Drag (C_D)');
ylabel('Coefficient of Lift (C_L)');
legend('Lift vs. Drag', sprintf('Max L/D Point (L/D = %.2f)', max_LD), 'Location',
'northwest');
set(gca, 'FontSize', 12);
saveas(gcf, "cl_vs_cd.png")
% --- Plot 2: Lift/Drag Ratio vs. Angle of Attack ---
figure;
plot(alpha_deg, L_over_D, 'b-', 'LineWidth', 2);
hold on;
plot(alpha_max_LD, max_LD, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); % Mark max
L/D
grid on;
title('Lift-to-Drag Ratio vs. Angle of Attack');
xlabel('Angle of Attack, \alpha (degrees)');
ylabel('Lift-to-Drag Ratio (L/D)');
legend('L/D Performance', sprintf('Peak Performance (L/D = %.2f)', max_LD), 'Location',
'southeast');
ylim([0, max_LD + 2]); % Set y-axis limits for better visualization
set(gca, 'FontSize', 12);
saveas(gcf, "lifttodrag.png")
```


$$V = \sqrt{\frac{2 * m * g}{\rho * S * C_L}}$$

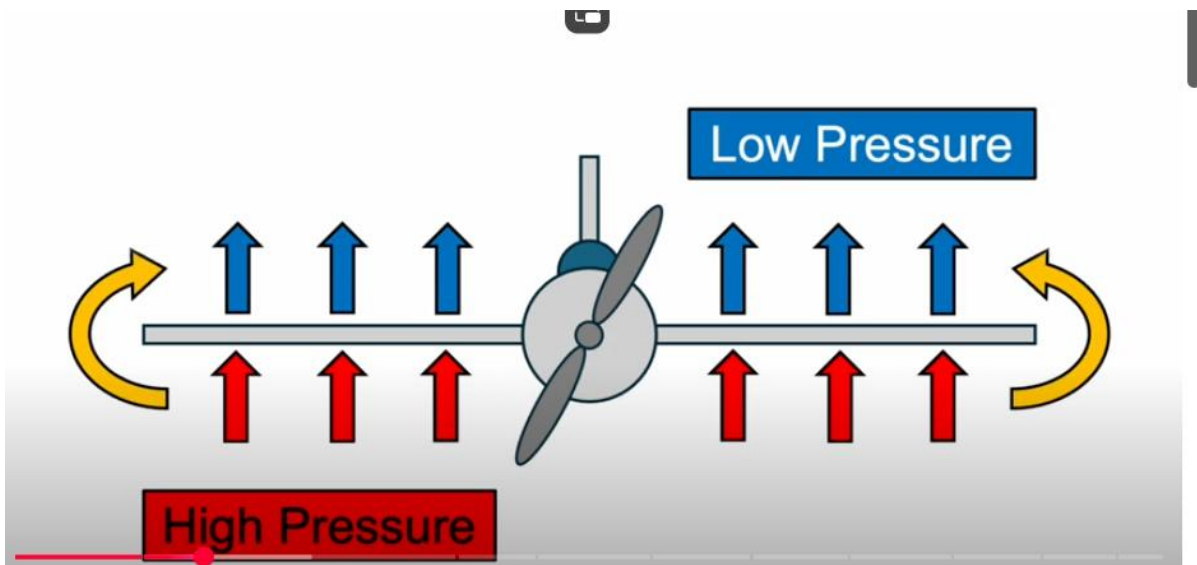
$$\frac{m}{S} = \text{wing loading}$$

Appendix C: Lift Coefficient For Different Angles of Attack and Cambers, and Diagram of The Turbulent Wake at a High Angle of Attack

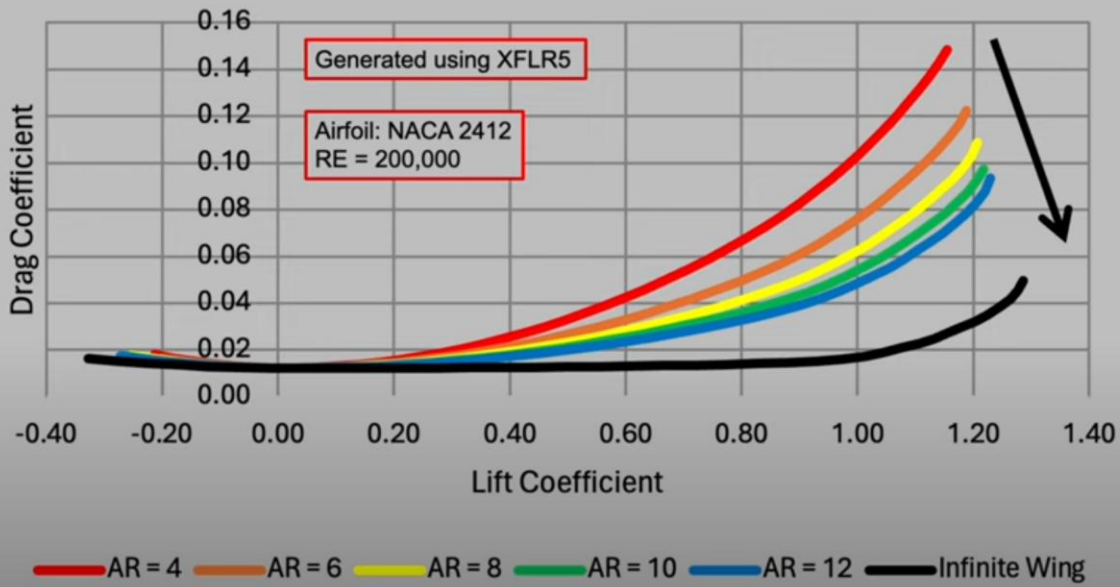




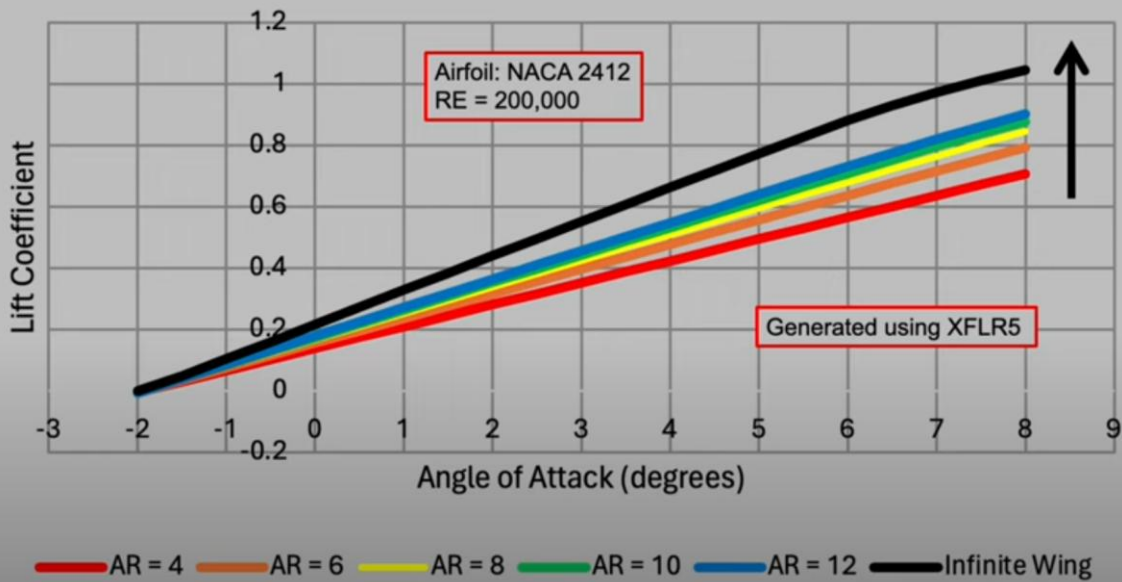
Appendix D: Diagram of Wingtip Vortices, and Graphs of The Aspect Ratio Performance Comparison For NACA 2414 Aerofoil



Aspect Ratio Comparison - Drag Polars



Aspect Ratio Comparison - Lift Curves



$$C_{d,i} = \frac{C_l^2}{\pi * AR * \overline{E}}$$

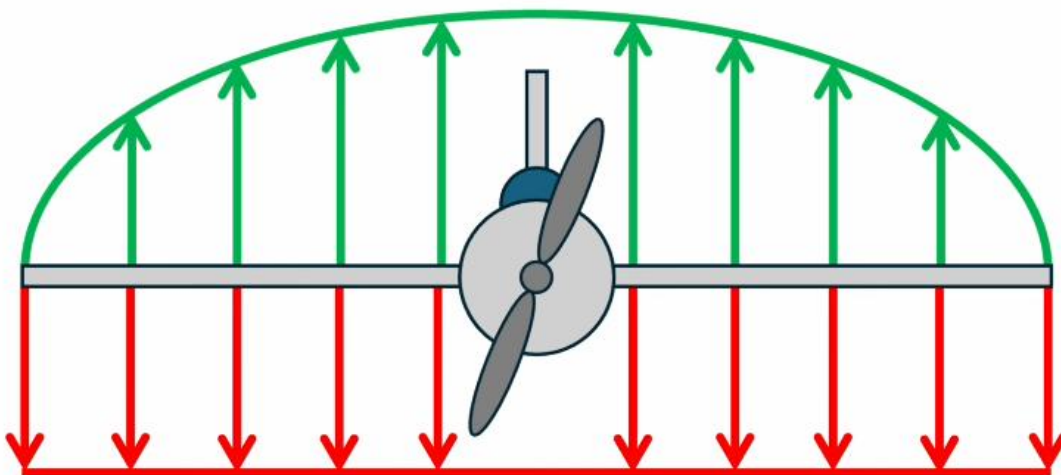
$C_{d,i}$ = Induced Drag Coefficient

C_l = Lift Coefficient

AR = Aspect Ratio

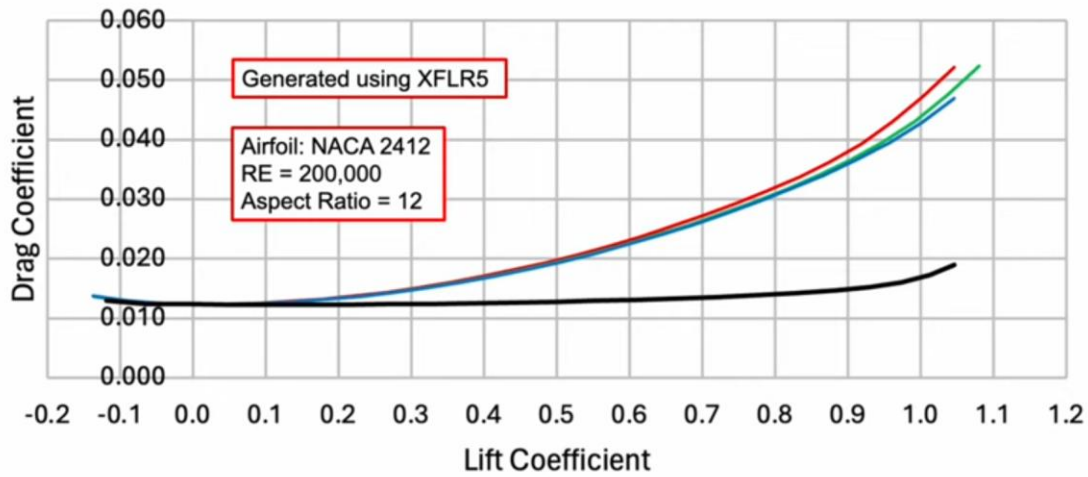
$\overline{E}$ = Oswald Span Efficiency Factor

Elliptical Lift Distribution



Constant Downwash Distribution

Tapered Wing Comparison - Drag Polars

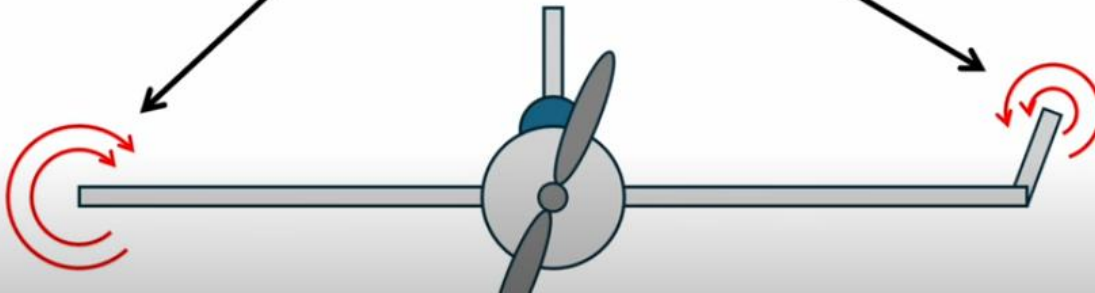


TR = 1.0 TR = 0.33 Elliptical Wing Infinite Wing

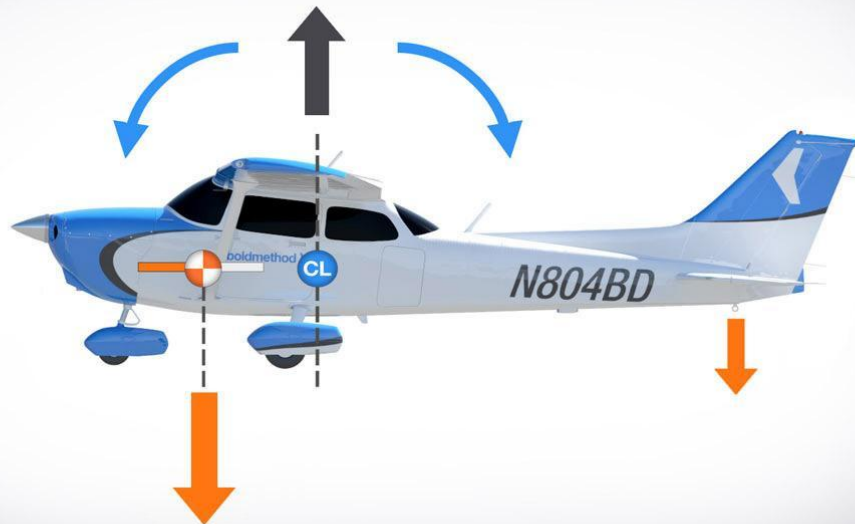


Large Vortex

Smaller Vortex



Tail Down Force Creates Moment In Opposite Direction



boldmethod

Appendix F Reynolds Number Initial Approximation

Reynolds Estimation - Based on Design Targets:

By DF 2025/17/10 \

$$Re = \frac{\rho v l}{\mu} = \frac{v l}{\nu}$$

Where:

- v = Velocity of the fluid
- l = The characteristics length, the chord width of an airfoil
- ρ = The density of the fluid
- μ = The dynamic viscosity of the fluid
- ν = The kinematic viscosity of the fluid

Instructions:

Estimate the Reynolds and Mach numbers that are appropriate to the flight of a glider that aligns with the design constraints of this project.

Constraints:

Constraint based :

- Wing Span = 1 M

Estimation based:

- Flight Speed Range: $v = 4-10$ m/s

Condition Assumptions:

- Assume Standard Sea Level Conditions **density** = 1.225 kg/m^3

[ISA tables tend to increment in a few thousand feet which isn't relevant to Monash's ~70m elevation]

- μ [Dynamic Viscosity of Air at this condition] = $1.8 \cdot 10^{-5} \text{ Pa}\cdot\text{s}$
- Temperature = 20 Degrees Celsius (293.15 K)
- V_{Sound} = 343 m/s
- Chord width – initially design for $1\text{m}/8 = 12.5 \text{ cm} / 0.125 \text{ m} / 125\text{mm}$

Okay So what do we have:

Density → Yep

Fluid Velocity Estimation → Yep

Length estimation → yes

Dynamic viscosity yes:

We now compute reynolds

```
In[1]:= (*Reynolds Number Computation*)
```

```
Reynoldsf[ $\rho$ _, v_, cw_,  $\mu$ _] := 
$$\frac{\rho * v * cw}{\mu}$$

```

```
In[6]:= reynoldsAt4mps = Reynoldsf[1.225, 4, 0.125,  $1.8 * 10^{-5}$ ]
```

```
reynoldsAt10mps = Reynoldsf[1.225, 10, 0.125,  $1.8 * 10^{-5}$ ]
```

```
Out[6]= 34027.8
```

```
Out[7]= 85069.4
```

| + |

Appendix G CAD